

Multi-Gov: Cross Chain Governance Audit Summary

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Version 2.0

Lead Auditors

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1 About Cyfrin

Cyfrin is a Web3 security company dedicated to bringing industry-leading protection and education to our partners and their projects. Our goal is to create a safe, reliable, and transparent environment for everyone in Web3 and DeFi. Learn more about us at cyfrin.io.

2 About This Document

This document is a public summary of the security audit Cyfrin performed for Wormhole Foundation on Multi-Gov: Cross Chain Governance. It records the repositories and commits reviewed, the timeline, auditors, in-scope files, and the number and final status of findings by severity. It closes with Cyfrin's attestation of the review's scope and final remediation status.

It is deliberately abbreviated: the technical detail of individual findings is not reproduced here. The complete audit report, version 2.0 dated October 4, 2024, was delivered to Wormhole Foundation, who retain it as the full record of the engagement. This summary was issued on October 29, 2026.

3 Disclaimer

The Cyfrin team makes every effort to find as many vulnerabilities in the code as possible in the given time but holds no responsibility for the findings in this document. A security audit by the team does not endorse the underlying business or product. The audit was time-boxed and the review of the code was solely on the security aspects of the in-scope code being audited. Unless the scope explicitly notes otherwise, the review is limited to the application layer and does not extend to the correctness of API/SDK/execution environment implementations, compilers, standard and third-party libraries, underlying operating systems and hardware.

4 Copyright and License

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5 Audit Overview

Over the course of 11 days, the Cyfrin team conducted an audit on the Multi-Gov: Cross Chain Governance code provided by Wormhole Foundation. The findings consist of 9 Low severity issues with the remainder being informational and gas optimizations.

Summary

Project Name	Multi-Gov: Cross Chain Governance
Repository	example-multigov
Commit	7513aa590008...
Audit Timeline	Aug 26th - Sep 9th, 2024
Methods	Manual Review, Stateful Fuzzing

Issues Found

Severity	Found	Resolved	Acknowledged
Critical Risk	0	0	0
High Risk	0	0	0
Medium Risk	0	0	0
Low Risk	9	4	5
Informational	6	3	3
Gas Optimizations	2	1	1
Total	17	8	9

6 Audit Scope

Cyfrin conducted a security audit of the MultiGov - Crosschain Governance system, focusing specifically on the EVM-compatible codebase. The audit was performed on the codebase at commit hash `7513aa5`.

This audit was limited only to the Solidity smart contracts and did not include any non-EVM contracts, off-chain components or front-end applications.

The following files in the `evm` folder were included in the scope of the audit:

```
src/HubEvmSpokeAggregateProposer.sol
src/HubEvmSpokeVoteDecoder.sol
src/HubGovernor.sol
src/HubMessageDispatcher.sol
src/HubProposalExtender.sol
src/HubProposalMetadata.sol
src/HubVotePool.sol
src/SpokeAirlock.sol
src/SpokeMessageExecutor.sol
src/SpokeMetadataCollector.sol
src/SpokeVoteAggregator.sol
src/WormholeDispatcher.sol
src/extensions/GovernorMinimumWeightedVoteWindow.sol
src/extensions/GovernorSettableFixedQuorum.sol
src/lib/Checkpoints.sol
src/lib/GovernorCountingFractional.sol
src/lib/SpokeCountingFractional.sol
src/interfaces/IHubVoteExtender.sol
src/interfaces/ISpokeVoteDecoder.sol
src/interfaces/IVoteExtender.sol
script/DeployHubContractsBaseImpl.s.sol
script/DeployHubContractsSepolia.s.sol
script/DeploySpokeContractsBaseImpl.sol
script/DeploySpokeContractsOptimismSepolia.sol
```

7 Risk Classification

	Impact: High	**Impact: Medium**	**Impact: Low**
Likelihood: High	Critical	High	Medium
Likelihood: Medium	High	Medium	Low
Likelihood: Low	Medium	Low	Low

8 Protocol Summary

MultiGov is a cross-chain governance system designed to enable decentralized decision-making across multiple blockchain networks. The protocol allows token holders to participate in governance activities, such as creating proposals and voting, regardless of the specific chain on which their tokens reside.

Key components of the MultiGov system include:

- **Hub chain:** Serves as the central coordination point for cross-chain governance activities. It hosts the main governance contract (HubGovernor) and manages the aggregation of votes from different chains.
- **Spoke chains:** Represent the various blockchain networks integrated into the MultiGov system. Each spoke chain contains contracts that facilitate local voting and communication with the hub.
- **Cross-chain communication:** Utilizes the Wormhole protocol to enable message passing between the hub and spoke chains.
- **Fractional voting:** Supports a voting mechanism where users can split their voting power across different options (For, Against, Abstain).

- **Vote aggregation:** Collects and combines votes from multiple chains.
- **Extensibility:** Includes features for extending proposal deadlines.

The protocol aims to solve the challenge of fragmented governance in multi-chain ecosystems, providing a unified decision-making platform for decentralized communities spread across different blockchain networks.

9 Attestation

Cyfrin attests that it performed a security review of the Multi-Gov: Cross Chain Governance code provided by Wormhole Foundation.

The repositories, commits, review period and findings with their final status are in the Summary and Issues Found tables in Section 5, and the in-scope files in Section 6. This attestation is limited to the documented scope, audited commits and mitigations specifically reviewed by Cyfrin.

Cyfrin publishes this document and its SHA-256 hash at https://github.com/Cyfrin/cyfrin-audit-reports. To check a copy on macOS or Linux, run the following in the folder that contains it; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name. Before running it, check that the URL is Cyfrin's repository.

```
U=https://github.com/Cyfrin/cyfrin-audit-reports/raw/main/reports/sha256
N=2024-10-04-cyfrin-wormhole-multigov-v2.0.pdf
F=$N
H=$(curl -fsSL $U/$N.sha256 | cut -d' ' -f1)
echo "$H *$F" | shasum -a 256 -c
```

Cyfrin's full audit report with all findings provided to Wormhole Foundation has the SHA-256 hash and file name below. To check a copy of it, first check this document as above, then run the following in the folder that contains the full report; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name.

```
H=e26a7aa6d7cf40b268fbd557ddb2a907765247ded898806071dd8261a1811eba
F=2024-10-04-cyfrin-wormhole-multigov-v2.0-full.pdf
echo "$H *$F" | shasum -a 256 -c
```

On Linux, `sha256sum -c` can be used in place of `shasum -a 256 -c`. On Windows, run `Get-FileHash <file> -Algorithm SHA256` in PowerShell and compare the result with the hash.